$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ldots + \frac{f^n(x^*)}{n!}(x-a)^n$$

where $x^* = a + \theta(x-a)$, $0 \le \theta \le 1$.

The above series is known as the Taylor series with Lagrange's form of the remainder (Here, remainder $R_n = \frac{f^n(x^*)}{n!}(x-a)^n$.We can have Taylor series with other form of remainder also. But for our purpose, it is sufficient to know this form of Taylor series).

It is also common to expand a function $f(x)$ about the point $x = 0$. The resulting series is described as **Maclaurin's series:**

$f(x) = f(0) + x f'(0) + (x^2/2!) f''(0) + \cdot\cdot\cdot + (x^n/n!) f^{(n)}(0) + \cdot\cdot\cdot$.

10.9 MEAN VALUE THEOREM AND L'HOPITAL'S RULE

If we put $n = 1$ in Taylor series, we get $f(x) = f(a) + R_0$

With suitable adjustment (a formal proof is beyond the scope of the course), we can derive that $R_0 = f'(a)(x-a)$.

This is known as *linearization of the function* $f(x)$ (as we are approximating the function f(x) by a polynomial of first degree).

This result is known as the Mean Value theorem. It states that if (i) $f(x)$ is continuous in the closed interval $a \le x \le b$ and (ii)$f'(x)$ exists in the open interval $a < x < b$, then there exists at least one value of x, c between a and b such that

$f(b) - f(a) = (b-a)\, f'(c)$.

Since,$a < c < b$, c can be written as $a + \theta(b-a)$, where $0 < \theta < 1$. Putting $b = a + h$, we get another form of the mean value theorem.

$f(a + h) = f(a) + hf'(a + \theta h)$, where $0 < \theta < 1$.

L'Hopital's Rule

L'Hopital's rule is another useful result (which can be proved by applying Taylor's theorem).

Suppose, $f(x) = g(x)/h(x)$. Now if both $g(x)$ and $h(x)$ approach either zero or infinity as $x \to a$, limit of the function $f(x)$ as $x \to a$ cannot be evaluated.

In that case, L'Hopital's rule states that $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)/h(x) = \lim_{x\to a} g'(x)/h'(x)$.

(If the latter limit exists).

Examples 10.9:

Using L'Hopital's rule, find

a) $\lim_{x\to 0}\frac{e^x-1}{x}$. b) $\lim_{x\to 0}\frac{(1+x)^n-1}{x}$

Here $g(x) = e^x - 1$ and $h(x) = x. g'(x) = e^x$ and $h'(x) = 1.$

Thus, $\lim_{x\to 0}\frac{e^x-1}{x}=\lim_{x\to 0}\frac{e^x}{1}=1$.

b) Here $g(x) = (1+x)^n-1$ and $h(x) = x. g'(x) = n(1+x)^{n-1}$ and $h'(x) = 1.$

Thus, $lim_{x\to 0}\frac{(1+x)^n-1}{x} = n$

Check Your Progress 2

1) Find Maclaurin's series for each of the following functions:

(i) y = e^x (ii) y = sin x (iii) y= cos x (iv) log (1+x)

....

....

....

....

2) Using L'Hopital's rule, evaluate each of the following:

(i) $\lim_{x\to 0}\frac{\log(1+x)}{x}$

(ii) $\lim_{x\to 1}\frac{x^3+3x-4}{2x^2+x-3}$

....

....

....

....

10.10 LET US SUM UP

In this unit we have discussed the basic techniques of differential calculus. We discussed various rules regarding how to differentiate different types of function. We applied this concept to illustrate average revenue, marginal revenue, average cost, marginal cost and elasticity of demand. We have also discussed slope and curvature of a function. Lastly, we have stated three important theorems Taylor series, mean value theorem and L'Hopital's rule.

10.11 KEY WORDS

Derivatives : The limiting value of the ratio of the change in a function to the corresponding change in its independent variable.

Differential Coefficient (or, **derivative**) : It refers to a measure of the rate of change of given function.

L'Hopital's Rule : Used to evaluate indeterminate forms of limits. It states that for indeterminate functions/forms (forms where the limits tend to 0/0 or + − ∞/∞), the limit of that form equals the limit of its derivatives.

Mean Value Theorem : If a function of one variable is continuous on a closed interval and differentiable on the interval minus its endpoints there is at least one point where the derivative of the function is equal to the slope of the line joining the endpoints of the curve representing the function on the interval.

Taylor Series : An infinite sum giving the value of a function *f(z)* in the neighbourhood of a point *a* in terms of the derivatives of the function evaluated at *a.*

10.12 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1)i) $\frac{dy}{dx} = 2\sin x \cos x - 2e^x + 4^x \log_e 4 = sin^2x - 2e^x + 4^x\log_e 4$

Note: Put $v = \sin x$. If $T = \sin^2 x = v^2$, then $\frac{dT}{dx} = \frac{dT}{dv} \cdot \frac{dv}{dx} = 2v.\cos x$

$= 2\sin x \cos x$. Thus, we are using the results II, III, VI to differentiate this simple looking expression.

ii) $\frac{dy}{dx} = 3(2x \operatorname{cosec} x - \operatorname{cosec} x \cot x) = 3 \operatorname{cosec} x\, (2x - \cot x)$

iii) Here, take $\phi(x) = logx - sinx$ and $\Psi(x) = 2x^2 - 1$. So,

$\phi'(x) = \frac{1}{x} - \cos x,\ \psi'(x) = 4x$. Thus,

$$\frac{dy}{dx} = \frac{(2x^2 - 1)(\frac{1}{x} - \cos x) - 4x(\log x - \sin x)}{(2x^2 - 1)^2}$$

$$= \frac{(2x - \frac{1}{x} - 2x^2 \cos x + 4x \sin x + \cos x - 4x \log x}{(2x^2 - 1)^2}$$

iv) $\frac{dy}{dv} = -2\sin v + 2\tan v \sec^2 v$ and $\frac{dv}{dx} = 3e^{3x}$.

Therefore,

$$\frac{dy}{dx} = \frac{dy}{dv}.\frac{dv}{dx} = 6e^{3x}[\tan(e^{3x})\sec^2(e^{3x}) - 2\sin(e^{3x})]$$

v) $\frac{dy}{dx} = \frac{dy}{d\phi}.\frac{d\phi}{dx} = -a\sin\phi.\frac{1}{b\cos\phi} = -\frac{a}{b}\tan\phi$

2)i) $dy/dx = -e^x + 4x^3$; $\frac{d^2y}{dx^2} = d(-e^x + 4x^3)/dx = -e^x + 12x^2$

$\frac{d^3y}{dx^3} = d\left(\frac{d^2y}{dx^2}\right)/dx = -e^x + 24x$.

ii) $d(3x^2\log(2x))/dx = 6x(\log(2x)) + 3x^2\{(1/2x)2\}$

$= 6x\log(2x) + 3x$

$\frac{d^2y}{dx^2} = d(6x\log(2x) + 3x)/dx = 6\{1.(\log 2x) + x(1/2x).2\} + 3$

$= 6\{(\log 2x) + 1\} + 9 = 6(\log 2x) + 15$

$\frac{d^3y}{dx^3} = d\left(\frac{d^2y}{dx^2}\right)/dx = d(6(\log 2x) + 15)/dx = 6\{(1/(2x)).2\} + 0 = 3/x$.

$\frac{d^3y}{dx^3} = d\left(\frac{d^2y}{dx^2}\right)/dx = d(3/x)/dx = 3(-1)/x^2 = -3/x^2$

3) (i) We know $d(e^x)/dx = e^x$; hence, $d^2(e^x)/dx^2 = d(e^x)/dx = e^x$

Continuing like this, we get $d^n(e^x)/dx^n = e^x$ for all n.

(ii) $y = \sin x$, $d(\sin x)/dx = \cos x$; $d^2(\sin x)/dx^2 = d(\cos x)/dx = -(\sin x)$

$d^3(\sin x)/dx^3 = d(-\sin x)/dx = -\cos x$

$d^4(\sin x)/dx^4 = d(-\cos x)/dx = \sin x$

Thus, we get

If $n = 4m$, then $d^n(\sin x)/dx^n = \sin x$

If $n = 4m + 1$, then $d^n(\sin x)/dx^n = \cos x$

If $n = 4m + 2$, then $d^n(\sin x)/dx^n = -(\sin x)$

If n =4m +3, then $d^n(\sin x)/dx^n = -\cos x$

(iii) y= cos x,

Repeating the process of (ii)above, we get

If n =4m+1, then $d^n(\cos x)/dx^n = -\sin x$

If n =4m +2, then $d^n(\cos x)/dx^n = -\cos x$

If n =4m +3, then $d^n(\sin x)/dx^n = (\sin x)$

If n =4m, then $d^n(\sin x)/dx^n = \cos x$

(iv) $y = 1/(1+x) = (1+x)^{-1}$

$d(1+x)^{-1})/dx = (-1)(1+x)^{-2}$

$d^2(1+x)^{-1})/dx^2 = (-1)(-2)(1+x)^{-3} = (2!)(1+x)^{-3}$

Similarly,

$d^3(1+x)^{-1})/dx^3 = (-3!)(1+x)^{-4}$

In general,

If n is even, then

$d^n(1+x)^{-1})/dx^n = (n!)[(1+x)^{-(n+1)}]$

If n is odd, then

$d^n(1+x)^{-1})/dx^n = (-n!)[(1+x)^{-(n+1)}]$

(v) y= log (1+x)

$d(\log(1+x))/dx = (1+x)^{-1}$

Then from (iv) above, we get

In general,

If n is odd, then

$d^n(\log(1+x))/dx^n = ((n-1)!)\ [(1+x)^{-n}]$

If n is even, then

$d^n(\log(1+x))/dx^n = (-\ (n-1)!)\ [(1+x)^{-n}]$.

Check Your Progress 2

1)(i)$d^n(e^x)/dx^n = e^x$ for all n.

Thus, $[d^n(e^x)/dx^n]_{x=0} = e^0 = 1$ for all n, … (a)

Using (a) in

Maclaurin's series expansion for $f(x) = e^x$

$f(x) = f(0) + x f'(0) + (x^2/2!) f''(0) + \cdot\cdot\cdot + (x^n/n!) f^{(n)}(0) + \cdot\cdot\cdot$., we get

$e^x = 1 + x + x^2 + \dots \dots + x^n + \dots \dots$

(ii)

$d^n(\text{Sin})/ dx^n$

If n =4m, then

$d^n(\sin 0)/ dx^n = \sin 0 = 0$

If n =4m +1, then

$d^n(\sin 0)/ dx^n = \cos 0 = 1$

If n =4m +2, then

$d^n(\sin 0)/ dx^n = -(\sin 0) = 0$

If n =4m +3, then

$d^n(\sin 0)/ dx^n = -\cos 0 = -1$

Maclaurin's series expansion for $f(x) = e^x$

$f(x) = f(0) + x f'(0) + (x^2/2!) f''(0) + \cdot\cdot\cdot + (x^n/n!) f^{(n)}(0) + \cdot\cdot\cdot .,$

we get

$\sin(x) = x - x^3/(3!) + x^5/(5!) - x^7/(7!) + \dots$

(iii) y= cos x

Following exactly the procedure of (ii) above, we get

$\cos(x) = 1 - x^2/(2!) + x^4/(4!) - x^6/(6!) + \dots$

(iv) log (1+x)

$d^n(\log(1+x))/ dx^n$ is given as follows

If n is odd, then

$d^n(\log(1+x))/ dx^n = ((n-1)!)\ [(1+x)^{-n}]$

Hence, $(d^n(\log(1+x))/ dx^n)_{x=0} = ((n-1)!)\ [(1+0)^{-n}] = (n-1)!$

If n is even, then

$d^n(\log(1+x))/ dx^n = (-\ (n-1)!)\ [(1+x)^{-n}]$.

Hence, $(d^n(\log(1+x))/ dx^n)_{x=0} = (-(n-1)!)\ [(1+0)^{-n}] = -(n-1)!$

For various constraints on the value of x and of $(1+x)^{-n}$, the Maclaurin's series is defined only for $-1 < x \leq 1$,

and using Maclaurin's expansion

$f(x) = f(0) + x\,f'(0) + (x^2/2!)\,f''(0) + \cdot\cdot\cdot + (x^n/n!)\,f^{(n)}(0) + \cdot\cdot\cdot\,.,$

we get

and is given by

$\log(1+x) = \log(1) + x\,(d(\log(1+x))/dx)_{x=0} + x^2\,(d^2(\log(1+x))/dx^2)_{x=0} + \ldots$

$= 0 + x\,(0!) + (x^2/2!)\,(-1!) + (x^3/3!)\,(2!) + \ldots$

$= x - x^2/2 + x^3/3 - x^4/4 + \ldots\ \ldots$

2. $(i) \lim_{x\to 0} \frac{\log(1+x)}{x} = \lim_{x\to 0} \frac{d(\log(1+x))/dx}{d\,(x0/dx} = \lim_{x\to 0} \frac{1/(1+x)}{1} = 1$

(ii) $\lim_{x\to 1} \frac{x^3+3x-4}{2x^2+x-3} = \lim_{x\to 1} \frac{3x^2+3}{4x+1} = \lim_{x\to 1} \frac{6x}{4} = \frac{3}{2}$

10.13 EXERCISES

Q1. The average cost (AC) of a firm is AC = $q^2 - 2q + 5$. The maximum capacity of the firm is 30 units. Find the range of the output for which AC is decreasing and for which it is increasing.

Ans. AC is decreasing when $\frac{d(AC)}{dq} < 0$. So $2q - 2 < 0\ and q < 1$, Thus, AC decreases for $0 < q < 1$ and AC increases for $1 < q < 30$.

Q2. The total cost function is given by $c = ae^{bq}$ (a, b are constants). Find the value of q for which marginal and average cost for this function is equal.

Ans. $AC = \frac{a}{q}e^{bq}$; $MC = \frac{dc}{dq} = bae^{bq}$ and

$AC = MC \rightarrow \frac{a}{q}e^{bq} = bae^{bq}$; i.e., $= \frac{1}{b}$.

Q3. The production function is given by $q = -\frac{L^3}{3} + 2L^2 + 12L$, where L is the amount of labour employed. Find the maximum amount of labour employed beyond which average return from labour starts diminishing.

Ans. Average return from labour $= \frac{q}{L} = -\frac{L^2}{3} + 2L + 12$; we have $\frac{d}{dL}\left(\frac{q}{L}\right) = -\frac{2L}{3} + 2$, when the above equals zero, we get a value of L below which it will turn negative, i.e., L = 3.

Q4. The demand curve for a consumer is α. k and α are constants. Find the price elasticity of demand.

Ans. –k log p = log α – log q, i.e., $-k\frac{dp}{p} = -\frac{dq}{q}$. Thus, $\varepsilon_d = \frac{dq}{q}\bigg/\frac{dp}{p} = k$

Q5. Total cost $c = 3q\left(\frac{q+7}{q+5}\right)+5$. Comment on the nature of MC.

Ans. $MC = \frac{dc}{dq}$. Then derive $\frac{dMC}{dq}$. Show that for continually right q, MC falls, i.e., $\frac{dMC}{dq} < 0$.

UNIT 11 DIFFERENTIAL CALCULUS: FUNCTIONS OF SEVERAL VARIABLES

Structure

11.0 OBJECTIVES

After going through this unit, you should be able to

- discuss the concept of differentiation;
- find out the partial derivatives of functions;
- determine the higher order derivatives and cross-partial derivatives;
- solve the problems based on partial derivatives;
- distinguish between total derivative and total differential; and
- identify homogenous functions with the help of their properties.

11.1 INTRODUCTION

This unit provides you with the detailed discussion and demonstration of the concepts of differential calculus. We start with the concept and definition of partial derivative. Further we try to understand the higher order derivatives and cross-partial derivatives. The unit has ample examples for better understanding of these concepts. Apart from reaching out to the difference between total derivative and total differential, we also demonstrate some applications of partial derivatives which we find in economics. We complete the unit by learning about differentiation of implicit functions and homogenous functions.

11.2 CONCEPT OF PARTIAL DIFFERENTIATION

So far, we have considered the functions of a single independent variable. However, most of the economic problems deal with the multivariate cases. For example, utility of a consumer depends on the consumption of multiple commodities and the production of a commodity depends on factors like land, labour, and capital and so on. Like this, other examples can be included to capture more interesting economic problems. Therefore, there is a need to analyse the concept of derivative for the class of functions with more than one independent variable.

For simplicity, we shall start with function of two independent variables. Let the function be $y = f(x_1, x_2)$. When we take such a function $y = f(x_1, x_2)$, with two independent variables, change in y can be examined as result of a change in either x_1 or x_2, or by both if we assume these to be independent. Thus, if we change only x_1, it will not affect x_2 but only y. The rate of change in y, assuming x_2 to remain constant, can be obtained by finding partial derivative of y with respect to x_1. Similarly, if x_2 alone is changed, the rate of change in y can be obtained from the partial derivative of y with respect to x_2.

Proceeding one-step further and assuming thatx_1 and x_2 are related to each other, if we change x_1, there will be change in x_2 (even when x_2 is not changed exogenously), as a result of the change in x_1. Thus, whenever we change the dependent variable y, it is affected in two ways: first, directly by the change in $x_1$and secondly, indirectly by the change in x_2 via the change in x_1. The rate of change of y due to a change in x_1 in this case is measured by total derivative of y with respect to x_1.

11.2.1 Partial Derivative Defined

The concept of 'Partial derivative' refers to the derivative of a multivariate function when only one of the independent variables is allowed to change, other variables remaining constant.